FS1
FIELD STUDY

OBSERVATIONS OF TEACHING -LEARNING IN ACTUAL SCHOOL ENVIRONMENT

BSED

FS 1

Observations of Teaching-Learning in Actual School Environment

FS 1
FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

Utilizing Teaching-Learning Resources and ICT

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

SPARK Your Interest

With the light speed by which technology is evolving now with 4.0, technology continues to be an ever significant part of the learning environment. This epsisode provides an opportunity for students to examine a Learning Resource Center or Multi-Media Center and learn about its collection, services, equipment, observe how a teacher utilies technology for instruction, and explore resources in the virtual learning environment. Students will analye and reflect on how technology, including artificial intelligence, supports the teaching-learning process.

TARGET Your Intended Learning Outcome

At the end of this Episode, I must be able to:

- Identify and classify learning resource materials in the multi-media center;
- Show skills in the positive use of ICT to facilitate the teaching-learning process (PPST 1.3.1);
- Show skills in the evaluation, selection, development, and use of a variety learning resources, including ICT to address learning goals (PPST 4.5.1);
- Analyze the level of technology integration in the classroom; and
- Demonstrate motivation to utilize ICT for professional development goals based on the PPST (PPST 7.5.1).

REVISIT the Learning Essentials

UNESCO ICT Competency Framework for Teachers Version 3 (ICT CFT v3, 2018)

The Information and Communications Technology Competency Framework for Teachers CFT) version 3 is a comprehensive framework guide teachers' development on the effective and appropriate use of ICT in education. It highlights what teachers should know and do clustered in six aspects, namely: 1. Understanding ICT in Education, 2. Curriculum and Assessment, 3. Pedagogy, 4. Application of Digital Skills, 5. Organization and administration, and 6. Teacher Professional Learning.

Similar to the PPST, the ICT CFT also articulated competencies in levels which guide teachers as they develop their ICT skills from Level 1, Knowledge Acquisition, to Level 2, Knowledge Deepening, and Level 3, Knowledge Creation. The Commission on Higher — Teacher Education Curricula (2017) includes the UNESCO ICT CFT so most likely you have tackled this in your Technology for Teaching and Learning classes.

It will be good to review the UNESCO ICT CFT v3 framework as you work in this episode. The activities here are meant for you to observe, analyze, and reflect about the competencies discussed in the framework. (Access it at https://www.open.edu/openlearncreate/pluginfile.php/306820/mod_resource/content/2/UNESCO%20ICT%20Competency%20Framework%20V3.pdf)

The Learning Resource Center

1. A school usually sets up a center that will provide valuable support to the teaching-learning process. Over the years the name of this center has evolved. Some of the names are Audiovisual Center, Media and Technology Resource Center, Teaching-Learning Technology Department, or Simply Learning Resource Center.

2. With the swift development of ICT, the natural outcome was the ever-expanding interface between the traditional library and ICT both in terms of hardware and software systems and applications.

3. Schools may have different set-ups when it comes to a Learning Resource Center (LRC). Some have replaced the term library with LRC. Some have a separate library, LRC, and Audio Visual or Media Center. Some only have the LRC both for teachers and students. Still, some have combined their learning resource centers with maker spaces.

4. The common: purpose among these centers is to provide print, audio-visual and ICT resources to support the teaching-learning process

5. The goals of the Center may include orienting and training teachers in the use of audiovisual and ICT resources, working with teachers and administrators in producing instructional materials, making available useful resources to the students, teachers, and the school community.

6. In order to support the philosophy and aims of the school, the Center must fulfill the following functions: center of resources, laboratory of learning, agent of teaching, service agency, coordinating agency, recreational reading center, and a link to other community resources.

Technology Integration

The Technology Integration Matrix provides a comprehensive framework for you to define and evaluate technology integration. It will provide you direction and guide you in the process of achieving effective teaching with technology. The teacher's integration of technology in instruction can be described as progressing in 5 levels: entry, adoption, adaptation, infusion and finally transformation.

The teacher also works at creating a learning environment that encourages and enables quality technology integration. The interdependent characteristics of the learning environment are being active, collaborative, constructive, authentic, and goal-directed.

The Technology Integration Matrix connects the Levels of Technology Integration and the Characteristics of the Learning Environment. Examine the matrix below. To make you understand how integration is done in each of the levels and environment, explore the TIM website and learn from the many interesting videos showing technology integration. Go to *http://fcit. usf.edu/matrix/matrix.php.*

Observing technology integration in the classroom

Technology Integration Matrix		**Levels of Technology Integration into the Curriculum**				
		Entry: The teacher uses technology to deliver curriculum content to students.	**Adoption:** The teacher directs students in the convention use of tool-based software. If such software is available, this level is the recommended entry point.	**Adaptation:** The teacher encourages adaptation of tool-based software to select a tool and modify its use to accomplish the task at hand.	**Infusion:** The teacher creates a learning environment that infuses the power of technology tools throughout the day and across subject areas.	**Transformation:** The teacher creates a rich learning environment in which students regularly engage in activities that would have been impossible to achieve without technology.
Characteristics of the Learning Environment	**ACTIVE:** Students are actively engaged in using technology as a tool rather than passively receiving information from the technology.	Students use technology for drill and practice and computer based training.	Students begin to utilize technology tools to create products, for example using a word processor to create a report.	Students have opportunities to select and modify technology tools to accomplish specific purpose, for example using colored cells on a spreadsheet to plan a garden.	Throughout the school day, students are empowered to select appropriate technology tools and actively apply them to the tasks at hand.	Given ongoing access to online resources, students actively select and pursue topics beyond the limitations of even the best school library.
	COLLABORATIVE: Students use technology tools to collaborate with others rather than working individually at all times.	Students primarily work alone when using technology.	Students have opportunities to utilize collaborative tools, such as email, in conventional ways.	Students have opportunities to select and modify technology tools to facilitate collaborative work.	Throughout the day and across subject areas, students utilize technology tools to facilitate collaborative learning.	Technology enables students to collaborate with peers and experts irrespective of time zone or physical distances.
	CONSTRUCTIVE: Students use technology tools to connect new	Technology is used to deliver	Students begin to utilize constructive tools	Students have opportunities to select and modify	Students utilize technology to make	Students use technology to construct , share,

	information to their prior knowledge rather than to passively receive information.	information to students.	such as graphics organizers to build upon prior knowledge and construct meaning.	technology tools to assist them in the construction of understanding.	connections and construct understanding across disciplines and throughout the day.	and publish knowledge to a worldwide audience.
	AUTHENTIC: Students use technology tools to link learning activities to the world beyond the instructional setting rather than working on decontextualized assignments.	Students use technology to complete assigned activities that are generally unrelated to real-world problems.	Students have opportunities to apply technology tools to some content-specific activities that are based on real world problem.	Students have opportunities to select and modify technology tools to solve problems based on real-world issues.	Students select appropriate technology tools to complete authentic tasks across disciplines.	By means of technology tools, students participate in outside-of-school projects and problem-solving activities that have meaning for the students and the community.
	GOAL-DIRECTED: Students use technology tools to set goals, plan activities, monitor progress, and evaluate results rather than simply completing assignments without reflection.	Students receive directions, guidance, and feedback from technology, rather than using technology tools to set goals, plan activities, monitor progress, or self-evaluate.	From time to time, students have opportunity to use technology to either plan, monitor, or evaluate an activity.	Students have opportunities to select and modify technology tools to facilitate goal-setting, planning, monitoring, and evaluating specific activities.	Students use technology tools to set goals, plan activities, monitor progress, and evaluate results throughout the curriculum.	Students engage in ongoing metacognitive activities at a level that would be unattainable without the support of technology tools.

Evaluation of ICT Resources

The world wide web is like an endless network of information, ever-expanding and almost limitless. Electronic resources come in different forms like websites, WebQuests, blogs, social network sites, on-line courses, a wide range of tools, and so many forms of apps.

As a future teacher, one of the skills that will be most useful for you is the ability not only to search for information but to make decisions, as to which ones you will take and use and which ones you will put aside. Aim to develop your skills in evaluating internet resources. You will be able to choose the best resources that will help you attain your teaching-learning objectives.

Below is a set of criteria which you can use to evaluate resources:

1. Accuracy. The resource material comes from a reliable source and is accurate, free from error and is up-to-date.
2. Appropriateness. The resource is grade/level-appropriate. The 'content matches what is needed by the teacher.
3. Clarity. The resource clearly addresses the instructional goals in mind.
4. Completeness. The content is complete. It has all the information needed to be able to use them.
5. Motivation. The resource is engaging and rewarding to learners. It will encourage the active participation of the learners.
6. Organization. The resource is logically sequenced. It clearly indicates which steps should be taken. The procedures or processes flow smoothly.

(Based on the work of Fitzgerald, Mary Ann, Lovin, Vicki, & Branch, Rober Maribe (2003). A Gateway to Educational Materials: An Evaluation of an Online Resource for Teachers and an Exploration of User Behaviors. Journal of Technology and Teache; Education. 11(1), 21-51).

Education 4.0

Shwab described the 4th wave of the industrial revolution. The unprecedented speed at which technology is evolving has disrupted many vital processes that involve how we run production, businesses and consequently how we teach and learn. **The following Technology trends have huge potential to transform the ways we teach and learn (UNESCO, 2018):**

1. **Open Educational Resources (OER).** OERs are materials that can be used for teaching and learning that do not require payment of royalties nor license fees. There is an abundance of OERs in the form of textbooks course materials, curriculum maps, streaming videos, multimedia apps, podcasts, and many others. They can have a significant impact on education as they are made available and easily accessible in the internet. Be sure to explore them to help you work on this episode.

2. **Social Networks**. Social networks have revolutionized the way we interact, learn about things and share information. Sites and apps such as Facebook, twitter, Instagram provide a virtual venue for teachers and learners to work together interact among themselves and with other classrooms locally and globally.

3. **Mobile Technologies.** Filipinos are one of the most active on the internet, and also one who sends the most number of text messages per day. This indicates the high number of mobile device users. These devices can also be used as a learning tool by allowing teachers and students more opportunities to learn inside and even outside the classroom.

4. **The Internet of Things**. IoT is a system of computing mechanisms that become built-in into many everyday things, that allow sending and receiving data through the internet. A lot of things have turned "smart." We have smart cars that can navigate on its ow2. Smart houses that monitor temperature and light. Smart TVs that interface with the internet. Watches that send our vital signs to our doctors. All of these developments can influence the way we teach and learn.

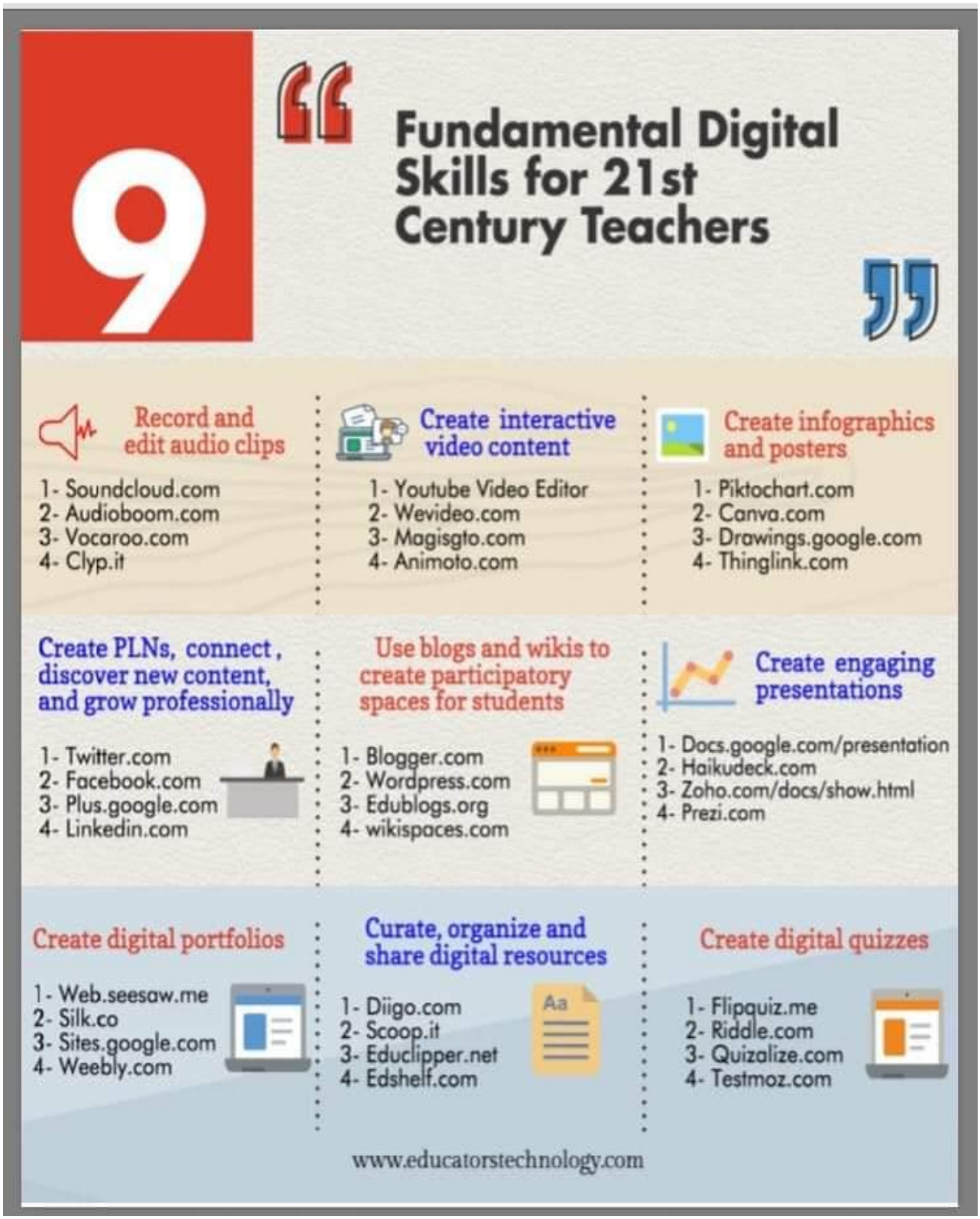
5. **Artificial Intelligence.** Commonly, artificial intelligence is associated when computers or machine simulate thinking and behaviors of humans such as talking, learning and solving problems. Virtual assistants such as Siri, Alexa, Bixby and Google assistant are near samples of AI. Among others, uses of AI in education can be in the areas of gamification and adaptive instruction for learners with special needs.

6. **Virtual Reality and Augmented Reality.** VR is a simulation of an environment by a computer program that allows a person to visit and experience the environment virtually. In AR, images from computer programs interlay with the actual views of the real-word, resulting in an extended, expanded, or altered view of a real-world environment. In education, among others, VR and AR programs and apps allow more exciting ways of seeing and experiencing things that add to the motivation and engagement of learners.

7. **Big Data**. Through the billions of interactions and transactions are done electronically, and through the internet, an enormous amount of data is generated and stored. The challenge is how to make sense of this data, through analytics and research, possibly answer pertinent questions about how to make teaching and learning most effective.

8. **Coding.** Coding is a skill necessary to create computer software, apps, and websites. Today, there are learning programs that introduce coding activities as early as kindergarten. Robotics programs in the elementary and secondary programs introduce and hone the skills of young learners. Coding helps learners develop novel ways of exploring and trying out ideas, especially when done with problem or project-based learning approaches.

9. **Ethics and privacy protection**. We have repeatedly heard about how quickly technology is changing and impacting our lives. Through all this. It is crucial to ensure that human Values and principles govern or guide our use of technology. Ethical practices that protect the rights of every person need to be upheld.


9
Fundamental Digital Skills for 21st Century Teachers
Record and edit audio clips
1- Soundcloud.com
2- Audioboom.com
3- Vocaroo.com
4- Clyp.it
Create interactive video content
1- Youtube Video Editor
2- Wevideo.com
3- Magisgto.com
4- Animoto.com
Create infographics and posters
1- Piktochart.com
2- Canva.com
3- Drawings.google.com
4- Thinglink.com
Create PLNs, connect, discover new content, and grow professionally
1- Twitter.com
2- Facebook.com
3- Plus.google.com
4- Linkedin.com
Use blogs and wikis to create participatory spaces for students
1- Blogger.com
2- Wordpress.com
3- Edublogs.org
4- wikispaces.com
Create engaging presentations
1- Docs.google.com/presentation
2- Haikudeck.com
3- Zoho.com/docs/show.html
4- Prezi.com
Create digital portfolios
1- Web.seesaw.me
2- Silk.co
3- Sites.google.com
4- Weebly.com
Curate, organize and share digital resources
1- Diigo.com
2- Scoop.it
3- Educlipper.net
4- Edshelf.com
Create digital quizzes
1- Flipquiz.me
2- Riddle.com
3- Quizalize.com
4- Testmoz.com
www.educatorstechnology.com

Massive Open On-line Courses

Massive Open On-line Courses (MOOCs) are online courses open to a big group of people. Below are the basic information about MOOCS. Let us focus on each word:

Massive

- MOOCs are on-line courses designed for large number of participants, usually larger than the number of students that can fit a regular classroom. There can be hundreds or even be a thousand students or more.

Open

- There is mostly freedom of place, pace, and time.
- Courses can be accessed by anyone anywhere as long as they have internet connection.
- Courses are open to everyone without entry qualifications. e Some courses are for free.

Online

- All aspects of the course are delivered online.

Course

The MOOC course offers a full course experience including:

- Educational content. May include video, audio, text, games, simulations, social media, and animation.
- Facilitation interaction among peers. Builds a learning community through opportunities to interact.
- Some interaction with the teacher or academic staff.
- Activities/tasks, tests, including feedback. Participants are provided with some feedback mechanism. Can be automatically generated like quizzes, feedback from peers or the teacher.
- Some kind of (non-formal) recognition options. It includes some kind of recognition like badges or certificate of completion. A formal certificate is optional and most likely has to be paid for.
- A study guide/syllabus. This includes instructions as to how you may learn from the materials and interactions presented.

  - Based on definition Massive Open Online Course v1.1 licensed under Creative Commons Attribution 4.0.

There are many MOOC providers like Edx, Coursera, Udacity, Udemy, Iversity. You will have a chance to explore them as you go through this episode.

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

OBSERVE, ANALYZE, REFLECT

Activity 11.1 Visiting the Learning Resource Center

Resource Teacher: **ROMMEL B. MANINANG JR.** Teacher's Signature:

School: **ABAÑO LABORATORY SCHOOL** Grade/Year Level: **9** Subject: **SCIENCE** Date:09/10/26

To realize the Intended Learning Outcomes, work through these steps:

1. Visit a school's Learning Resource Center. Look around and see what resources and facilities are available inside.

2. Ask the Learning Resource Center in charge about how some equipment or facilities are used.

3. Make an inventory of its available resources and classify them according to their characteristics and functions.

OBSERVE

As you visit and observe the Learning Resource Center, use the observation guide provided. Ask the assistance of the Center staff courteously.

An Observation Guide for a
LEARNING RESOURCE CENTER

Read the following statements carefully before you observe.

1. Go around the Learning Resource Center.
2. Find out what learning resources are present.
3. Examine and describe how the materials are arranged and how they are classified. Are they free from dust and moisture? Are they arranged for easy access?
4. Read the guidelines/procedures for borrowing materials. Are these guidelines? procedures posted are available for the users to refer to?
5. Familiarize yourself with the guidelines and procedures. Take photos of the center (if allowed).

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

After you are through with your observation, classify the resources available that you believe? are most useful. Use the **activity form** provided for you.

Name of Center Observed: **LIBRARY**
Date of Observation: **September 10, 2026**
Name of Observer: **JEWEL B. OCTA**
Course/Year/School: **BSED SCIENCES 4-A**

List of Available Learning Resources

Available Learning Resources (Enumerate in Bullet Form)	Characteristics and Unique Capabilities	Teaching Approaches where the Resource is Most Useful
1. Print Resources • Books • Journals • Research Papers	• Well-arranged and organized for accessibility • Serves as the primary resources for student learning	• Inquiry-Based Learning • Independent Learning • Collaborative Learning
2. Audio Resources • Laptop • Speaker	• Produces clear sound that supports library programs, orientation, faculty meeting, discussions, and other sessions.	• Experiential Learning • Collaborative Learning
3. Non-electronic Visual Resources • Posters • Bulletin Boards • White Board	• It serves as a long-term reminders that reinforce key points, rules, and study habits to guide students • It is also an alternative whenever electronic resources are currently unavailable	• Direct Instruction • Inquiry-Based Learning
1. ICT Resources • Internet • Televisiom • Computers	• Provides up-to-date and current information and resources • Allow access to multimedia resources • Support both independent and	• Experiential Learning • Collaborative Learning • Blended Learning • Inquiry-Based Learning • Technology-Based Learning

	dependent and collaborative tasks.	

Impression about the LRC:
The school provides almost everything that students meed to support their learning. It offers a clean and organized environment with proper lighting and ventilation, making it a comfortable place to study and learn.

Name and Signature of Observer:



JEWEL B. OCTA

Name and Signature of the Learning Resource Center In-charge:

ICT Competency Standards for Teachers (CHED Teacher Education Programs (2017):



Are the Learning resources/materials arranged properly according to their functions and characteristics?

- Yes, the learning materials are properly arrnaged according to their functions and characteristics. The resources are grouped into print resources, audio resources, nin-electronic visual resources, and ICT resources, making it easier to identify which amterial is appropriate for a particular learning activity. I also observed that each resources has a specific purpose. For example, books support independent and inquiry-based learning, while white board are useful for direct instruction and visual reinforcement. ICT resources such as the internet and muktimedia materials. This organized arrangement makes the leag resources center more useful and convenient for both teachers and students.

Do the guidelines and procedures facilitate easy access to the materials by the teachers? Why? Why not?

- Yes, the guidelines and procedures facilitate east access to the available materials because the resources are organized according to their type and purpose. Teachers can eaily determine which resources, from printed materials to ICT resources, also gives

teachers several options when preparing their lessons. However, accessibility can still depend on the availability of the equipments, internet connection, and whether certain resources are currently being used by others. Overall, the organized system hekps teachers save time and makes the process of finding appropriate learning materials easier.

What are the strengths of this Learning Resource Center?

- One of the main strengths of the learning resources center is thr variety if learning resources available. It provides print, audio, visual, and ICT resources that can support different teaching approaches and learning needs. I also consider its organization and accessibility as strengths because teacher and students can identify the appropriate materials more easily. The availability of ICT resources is another advantage because it allows access to updated information and multimedia learning materials. Hemce, the Learning Resource Center can support independent, collaborative, inquiry-based experiential, and technology-based learning.

What are the weaknesses?

- One weakness of Learning Resource Center is not all resources may always be readily available or accessible, especially ICT equipments that may be limited when several teachers or students need to use them at the same time. Some resources may also become outdated, particularly printed materials and technological equipments. Another possible weakness is the dependence on electricity and internet connectivity for ICT-based resources. If these are unavailable, teachers may need to rely on non-electronic resources instead. Hence, although the center has different types of materials, maintaining and updating them regularly is important.

What suggestions can you make?

- I would suggest that the Learning Resource Center should continue to update and expand its learning materials, particularly its digital and ICT resources. It would also be helpful to regularly check the condition of the computers, speakers , and other equipment to ensure that they are ready for use. The center could also provide clear labels, categories, or a simple borrowing and reservation system so teachers can easily locate and access the resources they need. I also suggest providing more current digutal materials and reliable internet access because these can support more interactive, collaborative, and technology-based learning experiences.



REFLECT

1. Which of the materials in the Learning Resource Center caught your interest the most? Why?

- The ICT resources, particularly computers and internet-based materials, caught my interest the most because they provide access to a wide variety of information and nu,timedia resources. As a future science teacher, I believe that technolog can make lessons more interactive and engaging, especially when teaching concepts that are difficult to understand through textbooks and discussions alone. Digital resources can provide videos, simulations, images, and other materials that allows students to visualize scientific concepts. I also find these resources useful because they can support both independent learning and collaborative activities.

2. Which gadgets/materials are you already confident to use/operate?

- I am already confident in using computers, laptops, speakers, the unetr et, and other basic ICT tools. I can use these resources to search for information, prepare presentations, create learning materials, show videos, and facilitate classroom activities. I am also comfortable using printed resources such as books when gathering information for lessons. These are tools that I have already encountered frequently during my studies, so I feel comfortable integrating them unti my future teaching.

3. Which ones do you feel you need to learn more about?

- I feel that I still need to learn more about advanced ICT tools and digital learning resources, particularly thise that can make classroom instruction more interactive. I want to improve my ability to use educational applications, simulations p, digital assessment tools, and other technology-based resources effectively. Knowing hiw to operate a gadget is different from knowing how to use it meaningfully for learning. As a future teacher, I want to develop the skills to select and use technology based on my students' needs and objectives of the lesson rather than simply using technology for sake of having it.

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

OBSERVE, ANALYZE, REFLECT

Activity 11.2 Observing Technology Integration in the Classroom

Resource Teacher: **ROMMEL B. MANINANG JR.** Teacher's Signature: 

School: **ABAÑO LABORATORY SCHOOL** Grade/Year Level: **9** Subject: **SCIENCE**
Date:**09/10/26**

To realize my Intended Learning Outcomes, I will work my way through these steps: Observe a class for three meetings. Video-tape, if allowed. **Step 2.** Describe how technology was integrated in the lessons and how the students were involved. **Step 3**. Use the Technology Integration Matrix to analyze the technology integration done by the teacher. **Step 4.** Reflect on what you have learned,



As you observe the class, use the observation sheets provided for you to document your observations.

Class Observation Guide

Read the following questions and instructions carefully before you observe.

1. What is the lesson about?
2. What visual aids/materials/learning resources is the teacher using?
3. Observe and take notes on how the teacher presents/uses the learning resources.
4. Closely observe the learners' response to the teacher's use of learning resources. Listen to their verbal responses. What do their responses indicate? Do their responses show attentiveness, eagerness, and understanding?
5. Focus on their non-verbal responses. Are they learning and are they showing their interest in the lesson and in the materials? Are they looking towards the direction of the teacher and the materials? Do their actions show attentiveness, eagerness, and understanding?

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

OBSERVATION SHEET NO. **REPORT** Date of Observation _____ School _____ Subject _____Topic _____
Grade/Year Level



UTILIZATION OF TEACHING AIDS FORM

Grade or year Leve; of Class Observed: **Grade 9**			
Date of Observation: **September 14, 2026**			
Subject Matter: **Science**			
Brief Description of Teaching Approach used by the Teacher: The teacher gave tge students the opportunity to discuss the lesson to their classmates . Through this approach, students were encouraged to actively participate, share their ideas, and learn from one another, making the discussion more interactive and student-centered.			
Teaching Aids used (Enumerate in Bullet Form)	**Strengths**	**Weaknesses**	**Appropriateness of the Teaching Aids used**
• Powerpoint Presentation	• Provides clear, well-organized flow of the lesson • Helps explain complex concepts by using	• Students may focus too much on slides instead of thinking critically • Can oversimplify lessons into	• Suitable for emphasizing key points and presenting examples. • Useful in illustrating processes clearly and making

	multimedia (text, bisuals, videos) • Students can easily follow step-by-step explanations.	short bullet points • Overuse may kimit interactions between students and teacher.	lessons more engaging. • Works best when slides are designed to support—not replace—teacher explanations.
• Laptop	• Allows the students to prepare structured and organized lessons. • Integrates technology smoothly into teaching • Supports various learning style through multimedia.	• Prone to technical issued such as software glitches or connectivity problems. • May reduce student engagement if the teacher relies too much on it.	• Best used for specific, purposeful learning tasks (e.g., research, collaborative activities, document editing) • Highly useful when incorporating specialized software in teaching

Use the Technology Integration Form to analyze the class you observed. Refer to the Technology Integration matrix on P. 4, In which level of technology integration do you think the teacher you observed operated? Why?

- I think the teacher operated at the Adoption level of technology integration. The teacher used a PowerPoint presentation and laptop to organize the lesson, explain complex concepts, and provide visual examples. The technology helped make the lesson clearer and more engaging, but it was mainly used to support the teacher's instruction rather than allowing students to independently use technology to create or explore new learning. However, the teacher still encouraged the students to discuss the lesson with their classmates, which made the lezrning interactive.

Based on the Technology Integration Matrix, what is the characteristics of the learning environment in the class you observed? Point your observations that justify your answer.

-
- The learning environment was interactive and somewhat student-centered because the studentswere guven the opportunity to discuss the lesson with their classmates and share their ideas. The powerpoint presentation helped provide a clear and organized flow of the lesson and supported students through visual examples and step-by-step explanations. However, since the technology was mostly used for presentating information, there was still room for greater student interaction with technology. The teacher could further encourage students to use digital tools for collaboration, exploration, and creating their own outputs.



1. Put yourself in place if the teacher. What would you do similarly and what would you do differently if you would teach the same lesson to the same group of students? Why?

- If I were the teacher, I would similarly use a powerpoint presentation and laptop because these tools help organize the lesson, emphasize important points, and explain concepts through visuals. I would also continue giving students opportunities to discuss the topic with their classmates because I believe students learn better when they can share ideas and learn from one another. However, I would do something differently by making use of technology more interactive and student-centered. Instead of relying mainly on slides, I would include short interactive activities, collaborative tasks, or digital exercises where students can actively participate and apply what they have learned. This would help prevent students from focusing too much on the slides and would make the technology a tool for learning rather than just a tool for presenting information.

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

OBSERVE, ANALYZE, REFLECT

Activity 11.3 **Exploring Education 4.0**

Resource Teacher: **ROMMEL B. MANINANG JR.** Teacher's Signature:

School: **ABAÑO LABORATORY SCHOOL** Grade/Year Level: **9** Subject: **SCIENCE** Date:09/10/26

OBSERVE

Class Observation Guide

Read the following statements carefully before you observe.

1. What is the lesson about? What are the teacher's objectives?

- The lesson is about the Mechanical Layers of the Earth's Interior. The teacher's objectives are for students differentiate the lithosphere, asthenosphere, mesosphere, outer core, and inner core based on their physical properties; classify Earth's mechanical layers according to their state and behavior; and appreciate how the properties of these layers help explain processes occuring inside Earth.

2. Note the important concepts that the teacher is emphasizing.
- The teacher is emphasizing the five mechanical layers of Earth—lithosphere, asthenosphere, mesosphere, outer core, and inner core—and their differences in terms of physical properties, state, and behavior. The lesson also highlights the distinction between compositional nd mechanical layers and how the characteristics of each layer are related to processes occuring within Earth.

3. Note the skills that the teacher is developing in the leaners.
- The teacher is developing students' skills in differentiating, classifying, comparing, and analyzing the mechanical layer of the Earth. Through the mechanical-layer diagram, visual aids, class discussion, and comparison activity, students are encouraged to identify similarities and differences among the layers and explain how their properties and behavior contribute to processes occuring inside the planet.



Analyzing the information, you got from observing the class, surf the internet to select electronic resources, including OERs, social networking sites, and apps with virtual or augmented reality ta=hat will be useful in teaching the same lesson. Evaluate the resources you found, using the set of criteria discussed in the Revisit the Learning Essentials part of this Episode. Use the form below to note your analysis and evaluation.

Electronic Resources Evaluation Form

Grade/Year Level	**GRADE 9**							
Subject Matter/Topic (Based on the class you observed)	**SCIENCE / Layers of the Earth**							
Lesson Objectives/Learning Outcomes	a) Differentiate the lithosphere, asthenosphere mesosphere, outer core, and inner core based on physical properties. b) Classify Earth's mechanical layers according to their state and behavior. c) Value how the properties of Earth's lavers help explain processes occurringinside the planet.							
Name and Type of Electronic Resources	Describe the electronic resource (include author/publisher/ source).	Put a check if the resource satisfies the criterion.						Describe how you can use it if you were to teach in the class you observed.
		Accurate	Appropriate	Clear	Complete	Motivating	Organized	
CANVA	It is an online design platform where you can create a powerpoint presentation, posters, and infographics.	/	/	/	/	/	/	It can be used to design creative powerpoint presentations about the layers of the Earth showing

								the difference between each layers including it's physical properties and other things.
GOOGLE DOCS	It is a web-based word processor where you can write, edit, and collaborate on documents in real-time.	/	/	/	/	/	/	It can be used for collaborative note-takinh where students can work together to summarize key points about the layers of the Earth.
YOUTUBE	It is a video sharing platform where you can watch and upload educational videos, documentaries, and animation.	/	/	/	/	/	/	It can be used to show educational videos or animations that explain the layers of the Earth.

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

ONLINE ARTICLES	A place of digital writing published on a website, blog, or digital magazine that you can read using a computer	/	/	/	/	/	/	Educational articles helps break down abstract planetary depths into clear, readable concepts.



REFLECT

1. Describe your experience in surfing the internet for appropriate electronic rec=sources for the class? What made is easy? Difficult?

- My experience in searching for electronic resources was generally easy and helpful because there are many available online materials that can support the lesson on the layers of the Earth. I was able to find resources such online articles that I cited for mg discussion. What made it easy was the accessibility if these resources and the variety of formats they offered (e.g., document). However, the difficulty oart was choosing resources that were accurate, clear, complete, and appropriate for my lesson. I had to carefully consider whether the information was reliable and whether the resources could actually help students understand the concepts I am discussing.

2. How did you choose which electronic resources to include here? What did you consider? Explain. Which of the new trends in Education 4.0 would you like to explore more for your work as a teacher? Why?

- I chose the electronic resources based on their accuracy, appropriateness, clarity, and organization, as reflected in the evaluation form. I considered whether each resources could effectively support the lesson objectives and make the topic easier for students to understand. For example, Canva can be used to create visual presentations. As a future teacher, I would like to expalore more of interactive and collaborative digital learning under Education 4.0. bevause I believe students learn better shen they are actively involved rather than simply listening to a lecture. These technologies can make science lesson nore engaging and accessible.

3. Reflect on your technology skills. What skills do you already have, and what skills would you continue to work on to be better at utilizing education 4.0 resources?

- I believe zi already have basic technology skills such as searching for information online, using Canva for making presentations and visual materials, encoding documents, and fact-checking informations. I am also comfortable using digital resources to organize and present information. However, I still need to improve my ability to evaluate online resources, create more interactive learning materials,ap and use digital tools creatively and effectively in teaching. I also want to become more confident in exploring new educational technologies so I can chiose resources that are not only attractive but also meaningful and useful for my students' learning.

OBSERVE, ANALYZE, REFLECT

Activity 11.4 **Professional Development Through MOOCS**

Resource Teacher: **ROMMEL B. MANINANG JR.** Teacher's Signature: 

School: **ABAÑO LABORATORY SCHOOL** Grade/Year Level: **9** Subject: **SCIENCE** Date: 09/10/2026

OBSERVE

To realize my **Intended Learning Outcomes**, I will work my way through these steps.

Step 1: Review the seven domains of PPST and identify competencies I like to develop more.

Step 2: Visit sites of MOOC providers and explore the courses offered that are relevant - to PPST domains I want to work on.

Step 3: Reflect on how I can continue developing my skills through MOOCS.

OBSERVE

I. Get a copy of the PPST and go over the competencies.

2. On the second column, write the competencies you like to work on.

3. Search for MOOCs in the internet which are relevant to the competencies you identified. You may try these sites:

- http://www.teachthought.com/technology/list-75-moocs-teachers-students/
- http://www.educationworld.com/a_curr/moocs-best-teachers-free-online-courses.
- html http://www.forbes. com/sites/skollworldforum/2013/06/10/moocs-forteachers-learners-too/

- https://www.mooc-list.com/categories/teacher-professional-development
- http://ww2.kqed. org/mindshift/2013/04/30/new-online-teacher-training-programjoins-mooc-madness/

4. Indicate the MOOC provider. You might need to create an account in the different MOOC providers to explore their MOOCs.

PPST Domain	Competencies I want to work on	MOOC related to the competency/ies (Include a short description)	MOOC Provider
1. Content Knowledge and Pedagogy	Integrate ICT tools to enhance teaching and learning.	"ICT in the Classroom" It helps teacher use technology to create an engaging, interactive, and meaningful lessons.	Coursera
2. The Learning Environment	Establish a positive and inclusive environment.	"Managing Behavior for Learning" It focuses on promoting a positive behavior and creating a supportive learning environment which can help the students focus more on the lesson.	FutureLearn
3. Diversity of Learners	Adapt teavhing methods to meet the needs of diverse learners.	"Introduction to Inclusive Education" It can hekp the teacher to design a lesson that will cater the diverse	Alison

		needs of the students.	
4. Curriculum and Planning	Strengthen kesson planning and curriculum alignment.	"Curriculum Design for Learning" It guides educators on how to align objectives, contents, and assessments for effective instruction.	EdX (University of London)
5. Assessing and Reporting	Develop fair and effective assessment tools.	"Assessment for Learning" Discusses different assessment types and how thry inform instruction and student feedback.	FutureLearn
6. Community Linkages and Professional Engagement	Collaborate with oarenfs and the local community for student success.	"Community Development" It introduces was educators can engage communities in supporting education intiatives.	TESDA Online Program
7. Personal Growth and Professional Development	Commit to continuous professional and reflection.	"Teacher Professional Development: Leading Change" It focuses on self-reflection, leadership,a nd	Coursera

		improvement as an educator.	



ANALYZE

From among the MOOCs you explored, pick at least three which you believe are the most appropriate for you. Describe the MOOCs below.

1. MOOC Title: **Integarte ICT Tool to Enhance Teaching and Learning**

Provider: **Coursera**

Objective of the MOOC:

To help teachers integrate technology effectively into their teaching practices in order to create engaging, interactive, and student-centered learning experiences. The course focuses on developing teachers' skills in selecting and applying appropriate ICT tools for instruction, collaboration, and assessment

Content Outline:

The course consists of modules focusing on:

- The role of ICT in 21st-century education and digital literacy
- Utilizing multimedia and online platforms to enhance lessons
- Designing interactive learning materials using ICT tools
- Integrating technology for formative and summative assessments

Why did you Pick this MOOC:

I picked this MOOC because I really want to improve the way I use technology in teaching. As a future science teacher, I believe ICT can make lessons more exciting and easier to understand, especially when explaining complex topics. I want to learn how to use different digital tools not just for presentations, but also to help my students participate actively and enjoy learning.

2. MOOC Title: **Adapt teaching methods to meet the needs of diverse learners**

Provider: **Alison**

Objective of the MOOC:

To equip teachers with the knowledge and skills to create inclusive classrooms that cater to the diverse learning needs of students. It aims to promote equality and ensure that every learner, regardless of background or ability, has access to quality education.

Content Outline:

This course is divided into several key topics, including:

- Understanding inclusion and learner diversity
- Addressing special educational needs and learning disabilities
- Differentiating instruction to accommodate all learners
- Fostering respect, empathy, and acceptance in the classroom
- Building lesson plans that promote participation and equity

Why did you Pick this MOOC:

I chose this course because I want to be a teacher who can understand and respond to the different needs of my students. I know that every learner is unique, so I want to learn how to make my lessons inclusive and fair for everyone. This course will help me build the right attitude and skills to make sure that all my students feel valued and capable of learning.

3. MOOC Title: **Establish a postive and inclusive classroom environment**

Provider: **FutureLearn**

Objective of the MOOC:

To develop teachers' understanding of classroom management and positive behavior strategies that contribute to a safe and supportive learning environment. It aims to enhance teachers' confidence in managing various student behaviors and maintaining an atmosphere conducive tolearning.

Content Outline:

The course is organized into the following modules:

- Understanding behavior and motivation in students
- Establishing classroom rules, routines, and expectations
- Promoting positive behavior through motivation and reinforcement
- Managing disruptions with empathy and consistency
- Building teacher-student relationships for mutual respect

Why did you Pick this MOOC:

I decided to take this MOOC because I know that managing student behavior is one of the most important skills a teacher must have. I want to learn how to handle different situations in the classroom calmly and

effectively. I also believe that creating a positive and respectful atmosphere will help my students feel safe and more focused on learning.



1. How can MOOCs help you in your future career as a professional teacher and as a lifelong learner?

- MOOCs can greatly support my career as a future professional teacher by providing flexible and accessible opportunities for continuous learning. They offer training in areas such as ICT integration, inclusive education, and effective assessment, which are essential in meeting the Philippine Professional Standards for Teachers (PPST). As a lifelong learner, MOOCs allow me to stay updated with new trends in education and develop innovative teaching practices. Through these courses, I can strengthen my competencies, improve my instructional strategies, and cultivate the habit of continuous professional growth, ensuring that I remain an effective and adaptive educator in an ever-changing educational landscape.

2. What did you learn from the way the providers use technology to teach in the MOOCs?

- I learned that MOOC providers effectively use technology to promote independent and flexible learning. They utilize various digital tools such as video lectures, interactive quizzes, and discussion forums that make the learning process engaging and student-centered. The way they organize modules and integrate multimedia elements shows how technology can support different learning styles and paces. This approach demonstrated to me that when technology is used meaningfully, it can create a more interactive and inclusive learning environment that encourages reflection, collaboration, and deeper understanding.

3. How will you prepare yourself for MOOCS, as a learner, and as a teacher who may someday teach a MOOC?

- To prepare myself for MOOCs by developing strong self-discipline, time management, and reflective learning habits to ensure that I can complete and apply what I learn from each course As a learner, I will actively engage in online discussions, explore digital tools, and apply new knowledge to my teaching practice. As a future teacher, I will study how MOOCs are structured and delivered, particularly how technology and interactive methods are used to enhance.

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT

SHOW Your Learning Artifacts

1. Include here pictures/illustration of the materials used by the teacher. Put your comments/ annotations about what you observed.



In order to assess what they have learned—the teacher gives the students tasks to create models to demonstrate what they have learned.



The teacher integrates ICT tools such as television, where he shows his Canva presentation in the class.

2. Visit www.teachnology.com or other teacher resource websites. Print useful instructional materials (worksheets, visual aids, flashcards, rubrics, etc.) and include them here. Indicate how they might be useful considering your major or area of specialization.

Name _____ Date _____

Color The Hemispheres!

Directions:

Color the North Hemisphere blue and the Southern Hemisphere red. Label the equator, north pole, and south pole.



3. Visit www.edudemic.com/50-educationtechnology-tools-every-teacher-should-know-about/

Explore and enjoy the fantastic education tools. Try them out. Describe what you discovered and share how these tools can be helpful to you. as a teacher

4. Visit edtechteacher.org. This a treasure box for you. Explore and share what you learned.
   - I Learned that EdTechTeacher,org is a valuable platform that helps teachers like me integrate technology more meaningfully in the classroom. It offers various resources such as workshops, webinars, and guides that focus on digital learning strategies like project-based learning and blended instruction. what l appreciate most is how it emphasizes using technology not just as a tool, but as a way to make learning more interactive, inclusive, and student-centered. Exploring this site made me realize how important it is to continuously grow as an educator and use technology to create engaging and effective learning experiences for all students.

5. Paste an article about an example of technology gadget/material that you want to learn more about. How can this gadget/material be useful in instruction/teaching?

- Interactive flat panel displays are large touchscreen displays designed specifically for modern cassrooms. They replace traditional whiteboards and projectors by offering multi-touch support, direct annotation, multimedia integration, and real-time collaboration tools. These panels enable teachers to display videos, simulations, diagrams, and interactive lessons on a single device—then invite students to come up and manipulate content, write responses, or demonstrate their thinking directly on the screen

Iwant to learn more about the Interactive Flat Panel Display (IFPD) because it offers a modern way to make teaching more engaging and interactive. This gadget can be very useful in instruction since it allows both teachers and students to interact directly with lesson materials, making learning more visual and hands-on. It helps students better understand complex concepts through videos, diagrams, and real-time collaboration. For teachers, it alen radıırae preparation time and replaces multiple tools with one digital device. I believe th 30/33 use lFPDs effectively will help me create a more dynamic, inclusive, and technological classroom environment

FS 1 FIELD STUDY

OBSERVATIONS OF TEACHING-LEARNING IN ACTUAL SCHOOL ENVIRONMENT

Episode 11 Utilizing Teaching-Learning Resources and ICT



EVALUATE Performance Task

Evaluate Your Work Task Field Study 1, **Episode 11-Utilizing Teaching-Learning Resources and ICT**
Learning Outcome: - Identify and classify learning resource materials in the multi-media center. - Show skills in the positive use of ICT to facilitate the teaching-learning process. - Show skills in the evaluation, selection, development, and use of a variety learning resources, including ICT to address learning goals. - Analyze the level of technology integration in the classroom. - Demonstrate motivation to utilize ICT for professional development goals based on the PPST.

Name of FS Student: **JEWEL B. OCTA** Date Submitted: **SEPTEMBER 15, 2026**
Year&Section: **BSED SCI 4-A** Course: **BSED SCIENCES**

LEARNING EPISODES	EXCELLENT 4	VERY SATISFACTORY 3	SATISFACTORY 2	NEEDS IMPROVEMENT 1
ACCOMPLISHED OBSERVATION SHEET	All observation questions/tasks completely answered/ accomplished.	One (1) or two (2) observation questions/tasks not answered/ accomplished	Three (3) observation questions/tasks not answered/ accomplished	Four (4) or more observation questions/ tasks not answered/ accomplished.
ANALYSIS	All questions were answered completely; answers are in depth and are thoroughly grounded on theories; grammar and spelling are free from error.	All questions were answered completely; answers are clearly connected to theories; grammar and spelling are free from errors.	Questions were not answered completely; answers are not clearly connected to theories; one (1) to three (3) grammatical spelling errors.	Four (4) or more observation were not answered; answers not connected to theories; more than four (4) grammatical/spelling errors.
REFLECTIONS	Profound and clear; supported by what were observed and analyzed	Clear but lacks depth; supported by what were observed and analyzed	Not so clear and shallow; somewhat supported by what were observed and analyzed	Unclear and shallow; rarely supported by what were observed and analyzed
LEARNING ARTIFACTS	Portfolio is reflected on the context of the learning outcomes; Complete, well-organized, highly relevant to the learning outcome	Portfolio is reflected on the context of the learning outcomes. Complete; well-organized, very relevant to the learning outcome	Portfolio is not reflected on in the context of the learning outcomes. Complete; not organized, relevant to the learning outcome	Portfolio is not reflected on in the context of the learning outcomes; not complete; not organized, not relevant
SUBMISSION	Submitted before the deadline	Submitted on deadline	Submitted a day after the deadline	Submitted two (2) days or more after the deadline
Comment/s				

SCORE	20	19-18	17	16	15	14	13-12	11	10	9-8	Below
GRADE	1.0	1.25	1.5	1.75	2.00	2.25	2.50	2.75	3.00	3.5	5.00
	99	96	93	90	87	84	81	78	75	72	71-Below

_____ _____
Signature of FS Teacher above Printed Name *Date*